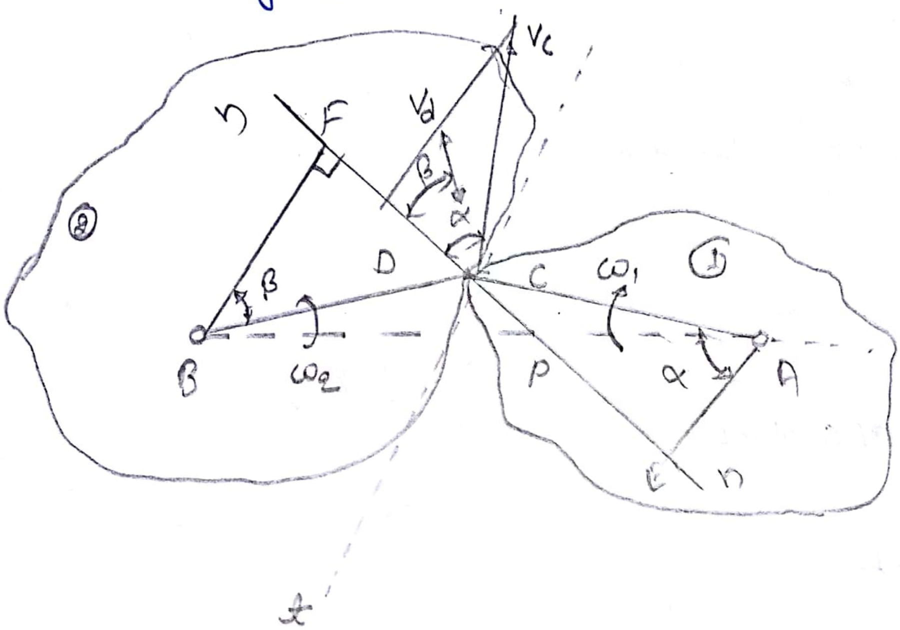
Law of gearing :-

The law of gearing states the condition which must be fulfilled by the gear tooth profiles to maintain a constant angular velocity ratio b/w two gears.



ω_1 = instantaneous angular velocity of the gear 1 (clockwise)

ω_2 = instantaneous angular velocity of the gear 2 (counter-clockwise)

V_c = linear velocity of C $= \omega_1 \cdot AC$

V_d = Linear velocity of D $= \omega_2 \cdot BD$

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19.) Face of teeth : Surface b/w pitch circle and the top land

20.) Flank of teeth : Surface b/w the pitch circle and the bottom land including fillet

21.) Top Land

22.) Face width

23.) profile

24.) Fillet radius

25.) Path of contact

*

26.) Length of path of contact

**

27.) Arc of contact

(a) Arc of approach

(b) Arc of recess

\# Contact Ratio = $\dfrac{\text{length of arc}}{\text{circular pitch}}$

$= \dfrac{\delta}{\gamma} = \dfrac{\alpha+\beta}{\gamma}$

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1) Pitch circle: imaginary circle ... by pure rolling action

2.) Pitch circle diameter :-

3.) pitch point : contact b/w two pitch circle

4.) pitch surface : surface of the rolling disc

5.) pressure angle or (φ) an angle of obliquity, angle b/w the common normal to two gear teeth at the point of contact and the common tangent at the pitch point.

pressure angle = $14\frac{1}{2}°$ and $20°$

6.) Addendum circle → circle passing through through the tips of teeth

7.) Dedendum or Root circle :→ circle passing through the roots of the teeth

8) Addendum :- It is the radial height of a tooth above the pitch circle. standard value one module

9.) Dedendum circle :-

10) (p) circular pith :- It is the distance measured along the circumference of the pitch circle from a point on one tooth to the corresponding point on the adjacent teeth ($p = \frac{\pi d}{T}$)

p → circular pitch
d = pitch dia
T = No. of teeth

11.) (P) Diametral pitch (not in SI unit) : It is the number of teeth per unit length of the pitch circle dia in inches ($P = \frac{T}{d}$)

12) Module : $m = \frac{d}{T}$ or $p = \pi \frac{d}{T} = \pi m$

13.) clearance :→ Radial difference b/w addendum and dedendum of a tooth
$= 1.157m - m = 0.157m$

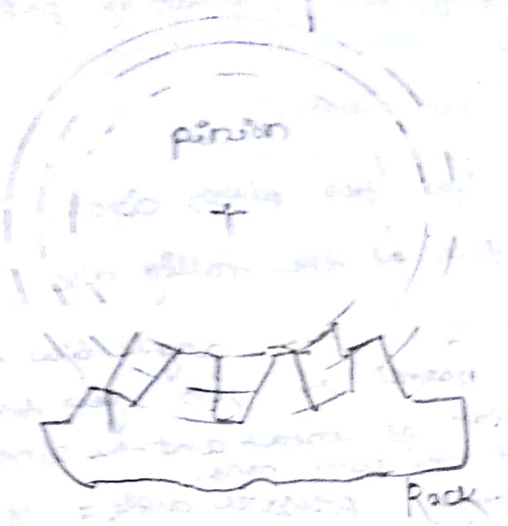
14) Total depth : Addendum + Dedendum

15) working depth = Sum of addendum of the two gears

16.) Tooth space

17.) Tooth thickness : It is the thickness of the t... measured along the pitch circle

18.) Backlash :- Space ...



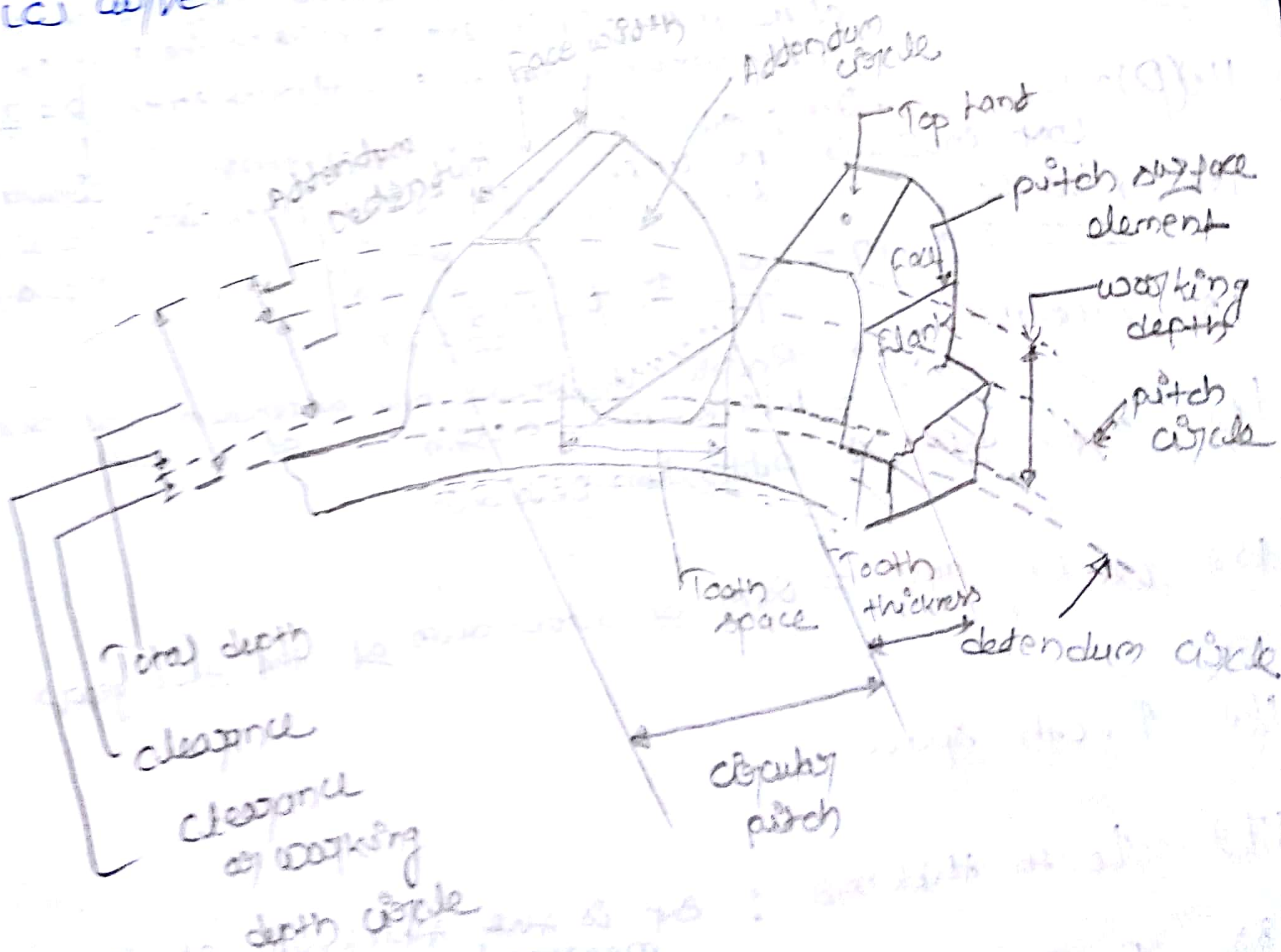
(a)

* (2) Acc. to positioning of teeth on the gear surface

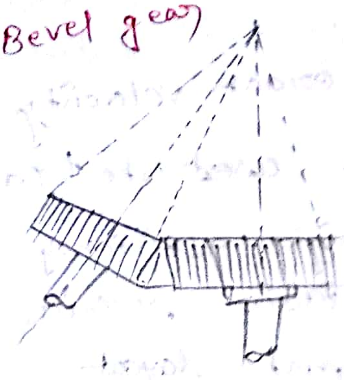
(a) straight

(b) inclined

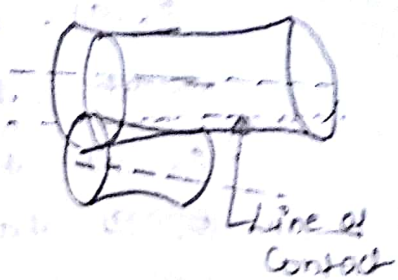
(c) curved



(c) Bevel gear



(d) spiral gear

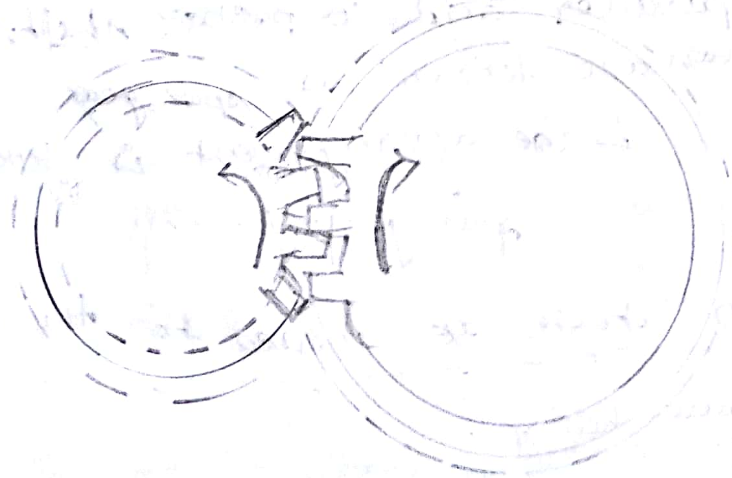


2. Acc. to peripheral velocity of the gears

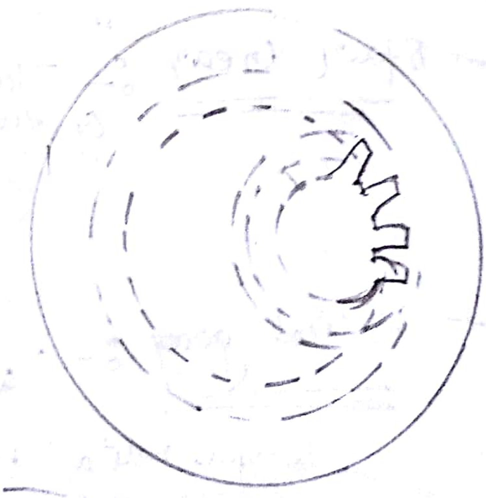
(a) Low velocity (velocity << 3m/s)
(b) Medium velocity (velocity b/w 3m/s - 15m/s)
(c) High velocity (velocity > 15m/s)

3. Acc. to type of gearing

(a) External gearing
(b) Internal gearing
(c) Rack and pinion



(a)



(b)

* Advantages and Disadvantages of Gear Drive

Advantages

1. It transmit exact velocity ratio
2. It may be used to transmit large power
3. It has high efficiency
4. It has reliable service
5. It has compact layout

Disadvantages

1. Require special tool and equipment
2. error in cutting teeth may cause vibrations and noise during operations.

* Classification of Toothed wheels / Gears

1. Acc. to the position of axes of the shaft

a) parallel (b) Intersecting (c) Non-intersecting and non-parallel

* Spur Gear :- Two parallel and co-planar shafts connected by the gears are known as spur gear & the arrangement is known as spur gearing.

* Helical gear :- when teeth are inclined to the axis.

(a) Single helical (b) Double helical

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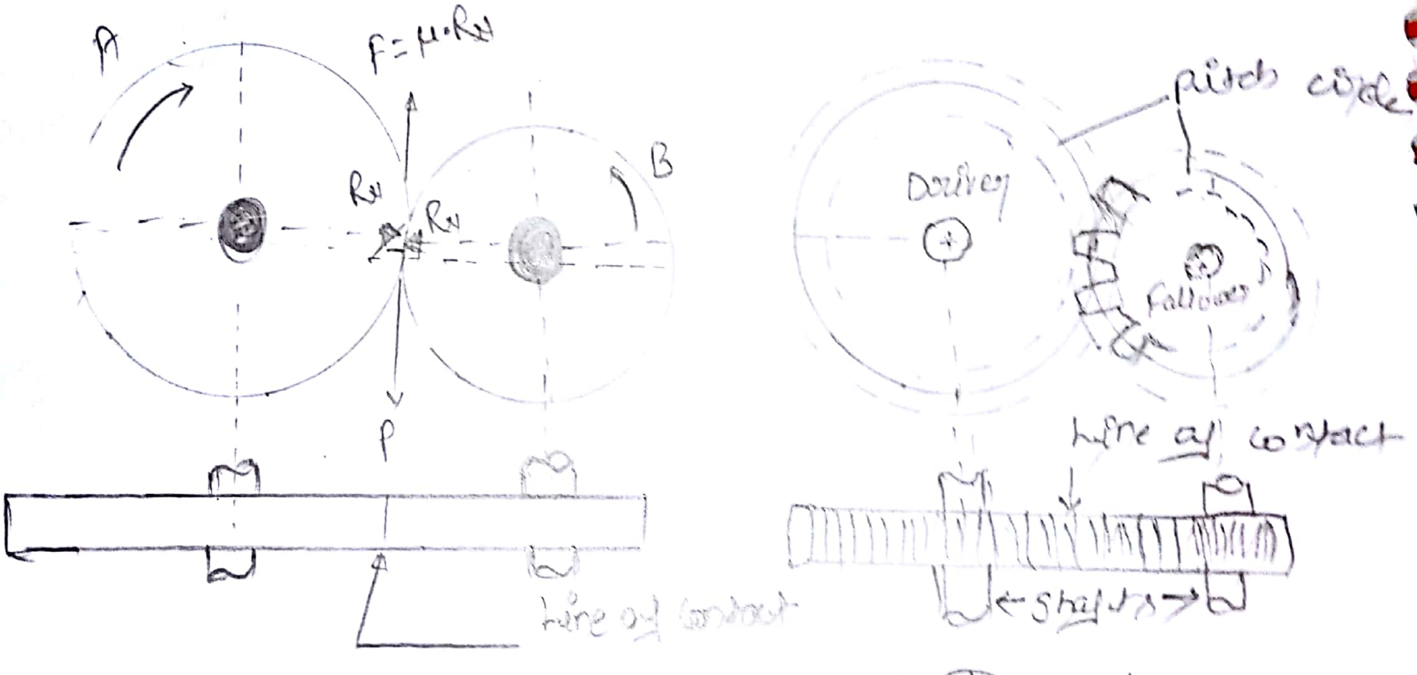
Unit:- III :> || Gears ||

{ Syllabus :- Geometry of tooth profiles, Law of gearing, Involute, Cycloidal, interference, helical, spiral }

Gears are used to transmit motion from one shaft to another or b/w a shaft and a slide { by engaging teeth }

* Friction wheels : The motion and power transmitted by gears is kinematically equivalent to that transmitted by friction wheels or disc



(a) Friction wheels

(b) Toothed gears

Fig (1)

P = Tangential force

F = frictional force

μ = Coefficient of friction

R_N = Normal reaction

" A friction wheel with the teeth cut on it is known as " toothed wheel or Gear "

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